

# Quantitative Research Notes

Finominal Quant Challenge

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## Executive Summary

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This project implements a fully reproducible monthly momentum strategy following the challenge specification.

The strategy ranks stocks using historical momentum, constructs a Top-10 portfolio using a momentum buffer rule, applies market-capitalization weighting with a 20% maximum position limit, and evaluates multiple momentum lookback windows.

Five lookback periods were tested (in trading days):

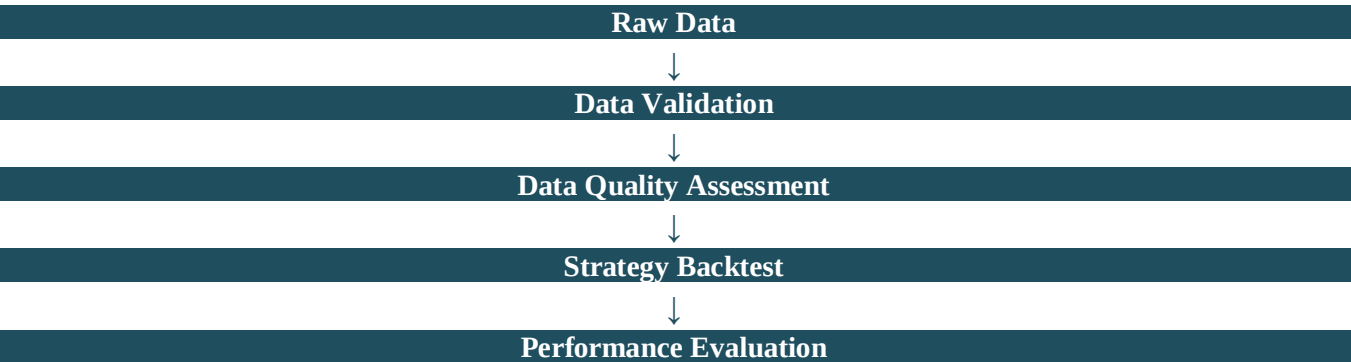
- 21
- 63
- 126
- 189
- 252

**The 126-day lookback produced the highest Annualised Sharpe Ratio among the tested lookback windows. A dedicated data-quality assessment was also conducted before backtesting, revealing numerous extreme price discontinuities in the supplied dataset that are likely to influence compounded return metrics. Consequently, the Sharpe Ratio is used as the primary optimisation criterion.**

## Research Workflow

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The complete research pipeline consists of five stages:



*This separation allows each stage to be verified independently while keeping the backtesting engine reproducible.*

## Methodology

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### Momentum Signal

Momentum is calculated using:

$$\text{Momentum} = \text{Price}(T\_calc) / \text{Price}(T\_calc - \text{Lookback}) - 1$$

where  $T\_calc$  is the trading day immediately before each monthly rebalance.

### Portfolio Construction

Each month, the following steps are applied:

- Rank all stocks by momentum.

- Select the highest-ranked stocks.
- Apply the momentum buffer rule: existing holdings are retained while their rank remains within the top 12.
- Remaining positions are filled from the highest-ranked stocks until the portfolio contains exactly ten securities.

This approach reduces unnecessary turnover.

## Portfolio Weighting

Weights are determined using market capitalization. Raw weights are computed from market capitalization on T\_calc.

An iterative redistribution algorithm enforces the 20% maximum position constraint while maintaining a fully invested portfolio.

## Backtesting

The strategy is rebalanced monthly beginning in January 2006. Portfolio returns are computed using daily returns between consecutive rebalance dates.

Performance is evaluated using:

- Annualized Sharpe Ratio
- Cumulative Return
- CAGR

## Software Engineering

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The implementation was designed using a modular architecture:

| Module              | Responsibility                              |
|---------------------|---------------------------------------------|
| src/data_loader.py  | Loads raw price and market-cap data         |
| src/validator.py    | Runs data validation checks                 |
| src/momentum.py     | Computes momentum signals                   |
| src/backtester.py   | Executes the monthly rebalancing backtest   |
| src/metrics.py      | Computes performance metrics                |
| src/optimizer.py    | Applies the weight redistribution algorithm |
| src/data_quality.py | Produces the data quality assessment        |

The project also includes independent unit tests covering:

- Validation
- Momentum
- Portfolio Construction
- Weight Calculation
- System Integration

This modular structure improves maintainability and reproducibility.

## Data Validation

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Before any calculations, the following validation checks are performed:

- ✓ Matching trading dates
- ✓ Matching stock universe
- ✓ Missing values
- ✓ Duplicate dates
- ✓ Positive prices
- ✓ Positive market capitalizations

**All validation checks passed successfully.**

## Data Quality Assessment

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A dedicated data quality module was developed to inspect the supplied dataset prior to backtesting. The report includes:

- Dataset summary
- Per-stock volatility statistics
- Extreme return detection
- CSV quality reports

The analysis identified numerous extreme one-day price changes, many substantially exceeding typical equity market behaviour.

The purpose of this assessment was diagnostic rather than corrective. The supplied dataset was intentionally left unchanged to preserve reproducibility and compliance with the challenge specification. In an institutional research environment, adjusted price series incorporating corporate actions would normally be used before factor construction and backtesting.

These observations are consistent with unadjusted corporate actions or vendor-specific data inconsistencies.

## Results

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**Among the evaluated lookback windows, 126 trading days produced the highest Annualised Sharpe Ratio.**

Because the supplied dataset contains multiple extreme price discontinuities, cumulative return and CAGR are heavily influenced by a small number of abnormal observations. Accordingly, these metrics should be interpreted with caution.

For model selection, the Annualised Sharpe Ratio was treated as the primary optimisation objective, consistent with the challenge specification.

## Limitations

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The supplied dataset does not contain adjusted prices or corporate action metadata. Consequently, the following may be affected by discontinuous price movements:

- Momentum rankings
- Daily returns

- Long-term compounded returns

No modifications were made to the supplied data. The objective of the project was to implement the required strategy exactly as specified while documenting any observed data-quality limitations. Any future deployment of the strategy should use split-adjusted and corporate-action-adjusted price data obtained from a reliable market data provider.

## Future Improvements

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Future work could include:

- Corporate-action-adjusted price series using vendor adjustment factors
- Transaction costs
- Slippage modelling
- Turnover analysis
- Maximum drawdown
- Information Ratio
- Rolling Sharpe Ratio
- Walk-forward optimization
- Portfolio turnover statistics
- Sector exposure analysis

## Conclusion

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The challenge specification was implemented exactly as prescribed.

The resulting framework is modular, reproducible, and fully tested.

The challenge specification was implemented exactly as prescribed. The resulting framework is modular, reproducible, fully tested, and optimised across multiple momentum lookback windows. In addition, a comprehensive data-quality assessment was performed to identify characteristics of the supplied dataset that may influence long-horizon performance metrics. These findings are documented to ensure transparent interpretation of the reported results.

### Appendix A — Data Quality Observations

- The supplied dataset exhibits a large number of extreme one-day price movements.
- A dedicated diagnostic module identified thousands of observations with unusually large daily returns, many exceeding typical equity market behaviour.
- These observations are consistent with unadjusted corporate actions or vendor-specific data inconsistencies.
- To preserve reproducibility and comply with the challenge specification, the supplied dataset was intentionally left unmodified.
- The data quality findings are provided for transparency and to support interpretation of the reported backtest results.